Vismay Vilas Chaudhari

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Education

Rochester Institute of Technology Aug 2022 - Dec 2025
Master of Science in Computer Science *Rochester, NY*

- **Coursework:** Data Structures & Algorithms, Object Oriented Programming, Machine Learning, Distributed Systems

Experience

Software Engineer Sept 2025 – Present
Ecolab *St. Paul, MN*

- Designed and maintained data pipelines processing data from 10+ upstream systems, ensuring reliable data flows across backend services and reporting workflows.
- Designed event-driven workflows using Azure Functions and Redis to automate data invalidation, reducing data freshness lag from hours to under a minute.
- Reduced query execution time by 30% by profiling and optimizing SQL transformations and join structures across large analytical datasets.
- Implemented observability and monitoring across production pipelines to detect data quality issues early and ensure system reliability.
- Built a validation suite covering business logic checks, null validation, outlier detection, and explosive value guards across production pipelines.

Software Engineering Co-op Aug 2024 – Aug 2025
Ecolab *St. Paul, MN*

- Processed large-scale datasets daily using batch workflows, supporting backend services and application features across multiple upstream data sources.
- Reduced pipeline execution time by 25% by profiling and eliminating inefficient query patterns across transformation workflows.
- Diagnosed and resolved production pipeline failures by tracing issues across ingestion, transformation, and inference stages.
- Developed .NET REST APIs to expose processed data for frontend applications, enabling consistent integration across reporting dashboards.

Software Engineer Oct 2020 – Jul 2022
Tata Consultancy Services *Mumbai, India*

- Developed backend services for real-time trading systems using ActiveMQ, handling high-throughput messaging and improving reliability across distributed components.

Projects

CodeCue – AI-Powered LeetCode Assistant | *React, TypeScript, Node.js, Express, LLaMA, Chrome Extension*

- Built a Chrome extension that integrates LLM-powered assistance directly into LeetCode, providing structured explanations and hints via LLaMA.
- Developed a React and TypeScript frontend with a Node.js backend, handling real-time prompt engineering and API response formatting.

Serverless Function Platform | *Python, Django, Docker, Kubernetes*

- Built a platform to execute user-submitted Python functions as isolated Kubernetes jobs with execution latency from 37ms to 3.5s.
- Added per-execution CPU and memory tracking to surface performance bottlenecks and debug inefficient workloads.

BugBlamer – CI/CD Debugging Assistant | *Python, FastAPI, LLaMA, Docker, GitLab API, GCP*

- Built an LLM-powered service that correlates Git diffs with CI log signals to identify the most likely commit responsible for build failures.
- Deployed a FastAPI backend with secure GitLab webhook integration via Docker on Cloud Run.

Valorant Strategy AI Chatbot – AWS x Riot Hackathon | *LangChain, OpenSearch, AWS Bedrock, Python*

- Built a RAG-based pipeline to analyze 5,000+ match reports using vector search with OpenSearch and retrieval flows orchestrated by LangChain.
- Integrated Bedrock-hosted LLMs to generate context-aware suggestions on AWS infrastructure.

Technical Skills

Languages: Python, Java, C#, JavaScript, TypeScript, SQL
Fullstack: React, FastAPI, Flask, Node.js, Express.js, Django
Cloud & Infrastructure: AWS (EC2, S3, Lambda), Azure, Docker, Kubernetes, CI/CD pipelines
AI & LLM: LLM APIs, RAG, embeddings, LangChain, AWS Bedrock
Databases: Snowflake, PostgreSQL, DynamoDB, Redis, MongoDB